Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the classification of various classes using k-Nearest Neighbor approach**	
Experiment No: 03	**Date:**	**Enrollment No: 92410133038**

Aim: To obtain the classification of various classes using k-Nearest Neighbor approach

IDE: Google Collab

Theory:

This algorithm is used to solve the classification model problems. K-nearest neighbor or K-NN algorithm basically creates an imaginary boundary to classify the data. When new data points come in, the algorithm will try to predict that to the nearest of the boundary line. Therefore, larger k value means smother curves of separation resulting in less complex models. Whereas, smaller k value tends to over fit the data and resulting in complex models. It's very important to have the right k-value when analyzing the dataset to avoid over fitting and under fitting of the dataset.

The model representation for KNN is the entire training dataset. It is as simple as that. KNN has no model other than storing the entire dataset, so there is no learning required. Efficient implementations can store the data using complex data structures like k-d trees to make look-up and matching of new patterns during prediction efficient. Because the entire training dataset is stored, you may want to think carefully about the consistency of your training data. It might be a good idea to curate it, update it often as new data becomes available and remove erroneous and outlier data.

Pre Lab Exercise:

A. What is K in KNN-approach?

In KNN (K-Nearest Neighbour), **K** represents the number of nearest neighbours considered while classifying a new data point. The algorithm checks the K closest data points and assigns the new point to the class which has the majority among those K neighbours

B. Which are the types of distance metrices?

1. Euclidean Distance
2. Manhattan Distance
3. Minkowski Distance
4. Hamming Distance (for categorical data.

c. What can be the criteria of selection of the value of K?

The value of K should not be too small or too large.

- A small K may cause overfitting.
- A large K may cause underfitting.

  Usually, K is selected by trying different values and choosing the one which gives better accuracy. Odd values are generally preferred to avoid tie situations.

d. What are the advantages of KNN algorithm?

1. Simple and easy to implement.
2. No training phase required.
3. Works well for small datasets.
4. Can be used for both classification and regression.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data
5. Visualize the Data

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6. Separate the feature and prediction value columns
7. Select the number K of the neighbors
8. Calculate the Euclidean distance of K number of neighbors
9. Take the K nearest neighbors as per the calculated Euclidean distance.
10. Among these k neighbors, count the number of the data points in each category.
11. Assign the new data points to that category for which the number of the neighbor is maximum.
12. Our model is ready.

Program (Code):

```
import numpy as np
```

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```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.preprocessing import StandardScaler

# Sample dataset (replace with your dataset)
from sklearn.datasets import load_iris
data = load_iris()
X = data.data
y = data.target

# Train test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Feature scaling
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

# Model
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
```

```
KNeighborsClassifier
KNeighborsClassifier()
```

```
# Prediction
y_pred = model.predict(X_test)

# Results
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("Accuracy Score:", accuracy_score(y_test, y_pred))
```

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```
Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
Accuracy Score: 1.0
```

Program(Code & Output):-

Untitled2.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[1]
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

[2]
```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
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[3]
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[4]
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Results:

a. **Confusion matrix**

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

b. **Accuracy Score**

1.0

Observation and Result Analysis:

a. Nature of the dataset

_ The dataset consists of multiple features and class labels. The data is suitable for classification as it contains distinct categories. The distribution of classes appears balanced

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b. During Training Process

Since KNN is a lazy learner, there is no actual training process. The algorithm simply stores the training dataset. Computation mainly happens during prediction

c. After the training Process

After prediction, the model classified the test data based on nearest neighbors. The confusion matrix shows most values along the diagonal, indicating correct classification.

d. Observation over the Learning Curve

_ As the value of K changes, the accuracy changes. Smaller K gives complex decision boundary, while larger K smoothens the boundary. Proper K value gives balanced performance.

Post Lab Exercise:

A. What are the major assumptions considered in linear regression

1. There should be a linear relationship between independent and dependent variables.
2. The errors should be independent.
3. The variance of errors should be constant (homoscedasticity).
4. The errors should be normally distributed (in many cases).

B. Why MSE is used instead of MAE for calculating the loss function

MSE (Mean Squared Error) is used because it penalizes larger errors more than smaller errors due to squaring. It is also mathematically convenient because it is differentiable, which makes gradient descent easier to apply. MAE is not as smooth as MSE, so optimization becomes slightly more difficult

C. How can the behaviour of outliers be understood while dealing with the unseen dataset

Outliers can be identified using visualization techniques like scatter plots or box plots. If some data points lie far away from the general pattern, they are considered outliers. Since linear regression is sensitive to outliers, they can significantly affect the slope of the line. Therefore, it is important to detect and handle them properly before training

D. Derive the Normal Equation for the Linear Regression.

For linear regression, the hypothesis is:

$y = X\theta$

The cost function is:

$J(\theta) = (1/2m)\,(X\theta - y)^2$

To minimize the cost function, we take the derivative with respect to θ and set it equal to zero.

After simplification, we get the Normal Equation:

$\theta = (X^T X)^{-1} X^T y$

This equation directly gives the optimal values of parameters without using gradient descent.

Conclusion:

In this experiment, Linear Regression was implemented to find the best fit line for the given dataset. The relationship between the feature and target variable was analyzed using scatter plots. The model was trained using gradient descent to minimize the cost function and obtain optimal values of slope and intercept. After training, the best fit line was plotted which represented the linear relationship between the variables. This experiment helped in understanding how linear regression works and how it can be used for prediction tasks.